Closing the Fidelity Gap: Real-Stack-Grounded Link Abstraction for 5G Digital Twins

***Abstract*—Network digital twins are essential for developing O-RAN rApps before deployment, yet open platforms force a trade-off among protocol-stack realism, site-specific propagation, network scale, and closed-loop control—none combines all four. We address the fidelity gap at its root: the link-to-system (L2S) abstraction. Instead of idealized SINR-to-throughput curves, our twin maps per-resource-element SINR to throughput through a curve *measured on OpenAirInterface's (OAI) real physical layer* (`nr_dlsim` with LDPC decoding), so network-scale KPIs inherit a real receiver's implementation losses. We quantify this gap: at 10 dB SINR, the OAI curve delivers 2.16 bits/s/Hz—38% below the Shannon bound (3.46 bits/s/Hz)—and its MCS steps are non-uniform, spanning up to 2.2 dB versus the $\sim$1 dB of an idealized staircase. The twin builds multi-cell, multi-user 5G scenes from OpenStreetMap, ray-traces every cell-to-user channel with Sionna RT, and hosts a proportional-fair traffic-steering rApp. Two bridges connect the analytical twin to a live OAI stack: exact ray-traced channel injection and a per-UE interference-as-noise-floor mapping. Across three cities and eight random layouts each (24 runs), the rApp raises median per-user throughput by $9.5 \pm 11.3\%$ and served cell-edge throughput by $22.4 \pm 24.8\%$, improves Jain fairness by 9.1%, and reduces peak cell load by 0.75 users, while leaving aggregate rate and coverage-limited outage unchanged. This gap is not cosmetic: replacing the OAI curve with an idealized one inflates throughput by 35% and changes the rApp's steering decisions in 42% of layouts while mis-estimating its per-deployment gain by a median of 7.5 points. Finally, we close the loop on a live OAI 5G Standalone stack—5G core, gNB, a real UE that registers and carries a PDU session, and iperf3 goodput over the UPF—on the same commodity host, and *measure* the fidelity gap end-to-end: at matched operating points the twin's link-level rate overpredicts deliverable goodput by a median of 53% (the MAC/TDD/transport overhead the link curve omits), and propagated through the rApp this mis-estimates the per-UE steering gain by 4.9 points. Even an OAI-grounded twin is thus optimistic until it is closed against the real stack.**

***Index Terms*—Digital twin, 5G NR, OpenAirInterface, link-to-system abstraction, O-RAN, rApp, traffic steering.**

I. Introduction

The digital twin is becoming the environment in which AI/ML-driven RAN control—O-RAN rApps and xApps—is trained and validated before it touches a production network [1]. Its value hinges on data fidelity: a traffic-steering rApp decides which cell serves each user from signal, interference, and load; those quantities depend jointly on 3D propagation (site-specific), on neighbouring cells (multi-cell), and on a link model that turns SINR into a real modulation-and-coding outcome (protocol-stack realism). Evaluating such a controller requires all of these together, and—because random deployments vary widely—across many layouts rather than a single scenario. A twin that reports a single headline number from one scenario can mislead: our own single-seed pilot suggested a uniformly large rApp gain, which a 24-run study tempered into a robust-direction, variable-magnitude result.

Open tooling splits into camps that do not meet. System-level simulators scale to many cells but abstract the physical (PHY) layer with idealized SINR-to-throughput curves. OAI-based emulators execute the real 5G-NR stack but, with realistic propagation, cover a single link. Commercial twins close the rApp loop but are proprietary. No open platform combines a real protocol stack, a site-specific ray-traced channel, a multi-cell network, and a closed control loop (Table I).

A subtler gap concerns the *L2S abstraction* itself. System-level evaluation cannot afford full PHY processing per user, so it relies on a mapping from SINR to block-error-rate (BLER). Open simulators use idealized or 3GPP-calibrated curves; we obtain ours from OAI's real receiver (`nr_dlsim` with LDPC decoding and rate matching), so network-scale KPIs inherit a real implementation's losses—particularly at MCS transition boundaries where control decisions are sensitive. At 10 dB SINR, the OAI-measured curve yields 2.16 bits/s/Hz, while the Shannon bound gives 3.46 bits/s/Hz—a 38% gap that an idealized curve would ignore (Fig. 2).

This paper contributes:

1) An *OAI-grounded L2S abstraction, and evidence that it matters for control.* The SINR-to-throughput curve is measured from OAI's real PHY (38% below Shannon at 10 dB, with non-uniform MCS steps up to 2.2 dB). We then show the fidelity gap is first-order, not cosmetic: replacing this curve with an idealized one changes the traffic-steering rApp's decisions in 42% of layouts and mis-estimates its per-deployment gain by a median of 7.5 points.
2) An open, CPU-only, multi-cell/multi-user digital twin: OpenStreetMap geometry $\rightarrow$ Sionna RT ray tracing $\rightarrow$ per-UE/cell throughput, with a swappable network source for real cell-site data.
3) Two twin-to-real-stack bridges: (a) exact ray-traced channel injection into a live OAI gNB–UE link, and (b) a per-UE interference-as-noise-floor mapping that lets a single-link OAI run reflect its multi-cell context.

4) *A measured, end-to-end fidelity gap.* We stand up the full OAI 5G Standalone loop—5G core, gNB, a real UE PDU session, and iperf3 goodput over the UPF—on the same commodity host, and measure deliverable goodput against the twin's prediction: the real stack delivers a median 53% less than the link-level rate across the MCS range, and the rApp's per-UE steering gain is mis-estimated by 4.9 ± 4.3 points. This makes concrete the difference between what the twin predicts and what the real stack delivers.
5) A rigorous multi-scene, multi-seed evaluation (24 runs across 3 cities) of a proportional-fair traffic-steering rApp, with honest reporting of variance and outage, a load-sensitivity sweep separating coverage-limited and interference-limited regimes, and a correlation analysis showing rApp gain is limited by baseline outage rather than by raw load imbalance.

II. Related Work

Open RAN digital-twin platforms. Table I positions the closest open efforts on four axes. OWDT [2] pairs OAI with Sionna RT for a faithful single gNB–UE link but is single-cell and monitor-only. Tiny-Twin [3] runs a real OAI stack for multiple UEs with a RIC loop, but single-cell from replayed channel traces. Orange's RIC-TaaP/ns-O-RAN and OpenTwin [4] close rApp loops across many cells—OpenTwin adds twin-to-real KPM self-correction—but on ns-3's abstracted PHY. Recent OAI+FlexRIC work [5] closes handover loops across real cells but over the air, without a site-specific ray-traced channel. Within OAI, an emerging ray-tracing channel emulator feeds the virtual-time simulator VRTSim (multi-node on its roadmap). Commercial twins (e.g., Rimedo Labs' NDT with VIAVI) close the loop but are neither open nor reproducible.

Link-to-system abstraction. System-level simulation [6] relies on L2S mapping: a link-level simulator characterises SINR-to-BLER for each MCS, and an effective-SINR mapping (EESM [7], [8]) collapses a frequency-selective SINR vector into an AWGN-equivalent value. The distinction here is that our lookup is measured from OAI's own PHY rather than from idealised curves, so network-scale KPIs inherit a real receiver's implementation losses. Our multi-cell interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [9], which loads unused resources to emulate other-cell activity; we instead derive each UE's interference from the ray-traced neighbour channels.

O-RAN control loops. The O-RAN architecture [10], [11] externalises RAN control: xApps on the near-RT RIC (tens of ms, via E2) and rApps on the non-RT RIC (seconds and above, via O1/A1). Traffic steering via cell-individual offsets (CIO) is a canonical rApp use case [1]. A digital twin lets such an rApp be trained and validated before deployment.

TABLE I
Open RAN digital-twin platforms on four axes.

Platform	Real stack	Site RT	Multi cell	Closed loop
OWDT [2]	✓	✓	×	×
Tiny-Twin [3]	✓	×[a]	×	✓
Orange ns-O-RAN / RIC-TaaP	×[b]	✓	✓	✓
OpenTwin [4]	×[b]	×	✓	✓
OAI+FlexRIC [5]	✓	×[c]	✓	✓
OAI VRTSim / RT emulator	✓	✓	×[d]	×
This work	✓	✓	✓	✓*

[a]replayed traces; [b]ns-3 abstract PHY; [c]over-the-air; [d]multi-node on roadmap; *SA loop closed & measured; multi-cell steering grounded per-link.

III. Digital Twin Architecture

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer that covers every cell and UE, and a real-stack layer that grounds sampled links in OAI. Both are driven by the same ray-traced channel, and the network data source is swappable so that synthetic and real deployments run through an identical path.

Geometry. A WGS84 bounding box drives an OpenStreetMap (Overpass) query; footprints are extruded into a 3D Mitsuba scene with ITU materials.

Network. Cells and UEs come from a swappable source: a seeded random generator (N sites $\times$ 3 sectors, uniform UE drops) for controlled experiments, or a CSV reader for real cell-site coordinates (e.g., OpenCelliD). Both feed the same pipeline, so the twin can be calibrated on synthetic data and run on real site data without code changes.

Channel. Sionna RT [12] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE and computes the channel frequency response for every cell-to-UE link over an OFDM resource grid, capturing reflections, diffractions, and scattering from the building geometry. Ray-tracing depth is 3, balancing fidelity against CPU runtime.

IV. OAI-Grounded KPI Engine

The KPI engine is where the twin's fidelity is grounded. Let $\mathbf{H}_{u,c}(k) \in \mathbb{C}^{N_r \times N_t}$ be the ray-traced channel of link $c \to u$ on resource element (RE) k, and P_c^{tx} the cell transmit power. The wideband received power is $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \, \overline{\|\mathbf{H}_{u,c}\|_F^2}$, where the bar denotes averaging over antennas and REs. Each UE associates to the cell maximising received power biased by a cell-individual offset (CIO), the O-RAN control knob:

$$s(u) = \arg\max_c \; P_{u,c}^{\text{rx}} \cdot 10^{\,\text{CIO}_c/10}. \tag{1}$$

Because each cell serves one UE at a time under airtime sharing, the optimal single-stream precoder is the matched filter, so the serving per-RE gain is the largest singular value $\sigma_{\max}$ of the serving channel. With flat per-RE transmit power p_c, a neighbour load factor $\rho \in [0,1]$, and thermal noise $N_0 = kTB \cdot \text{NF}$, the per-RE SINR is

$$\gamma_u(k) = \frac{p_s \, \sigma_{\max}\big(\mathbf{H}_{u,s}(k)\big)^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \tag{2}$$

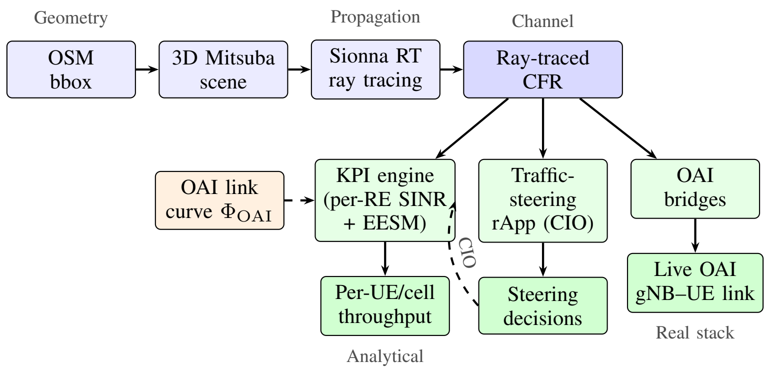

Fig. 1. Architecture. OpenStreetMap geometry and a swappable network source feed Sionna RT; the ray-traced channel drives (left) an analytical multi-cell KPI engine and the traffic-steering rApp, and (right) a live OAI link over the exact channel with a per-UE interference noise floor. The OAI-measured link curve Φ_{OAI} grounds the analytical throughput in a real receiver.

The per-RE SINRs are compressed to one effective SINR by exponential effective SINR mapping (EESM), which preserves the frequency selectivity the ray tracer produces:

$$\gamma_u^{\mathrm{eff}} = -\beta \ln\Big(\tfrac{1}{K}\sum_k e^{-\gamma_u(k)/\beta}\Big), \tag{3}$$

with per-modulation β calibratable by a single scale factor. The EESM β values are set per modulation order (QPSK, 16QAM, 64QAM) following established link-abstraction literature [8], and a single `eesm_beta_scale` parameter allows calibration against OAI in-the-loop ground truth.

OAI-measured link curve. The effective SINR is mapped to a modulation-and-coding scheme (MCS) and spectral efficiency η_u through a curve Φ_{OAI} *measured offline from OAI's* `nr_dlsim` (real LDPC decoding and rate matching), i.e. $(m_u, \eta_u) = \Phi_{\mathrm{OAI}}(\gamma_u^{\mathrm{eff}})$. This is the key distinction from prior system-level twins: instead of an idealised staircase or a 3GPP-calibrated analytical expression, the SINR-to-throughput mapping carries a real receiver's implementation losses—including LDPC decoding margins, rate-matching overhead, and the non-uniform SINR spacing between MCS transitions that a real stack exhibits. The curve is measured once by sweeping SNR $\times$ MCS through `nr_dlsim` and recording the lowest SNR at which first-transmission BLER stays within the target (10%); the resulting staircase (Fig. 2) is a monotone lookup table indexed by effective SINR.

Quantifying the implementation loss. Fig. 2 compares the OAI-measured curve against the Shannon bound and an idealized uniform-staircase. At 10 dB SINR, the OAI curve delivers 2.16 bits/s/Hz—38% below Shannon (3.46 bits/s/Hz). In relative terms the loss is largest at low SINR and narrows toward high SINR, but in absolute terms it grows to $\sim$1.6 bits/s/Hz at 64QAM, as LDPC and rate-matching overhead accumulate. Critically, the MCS transition spacing is non-uniform: the QPSK$\rightarrow$16QAM boundary spans 2.2 dB, while adjacent QPSK steps are $\sim$1 dB. An idealized staircase with uniform 1 dB steps would misplace this transition by over 1 dB, causing a system-level twin to assign the wrong MCS to boundary users—exactly the users a traffic-steering rApp aims to help.

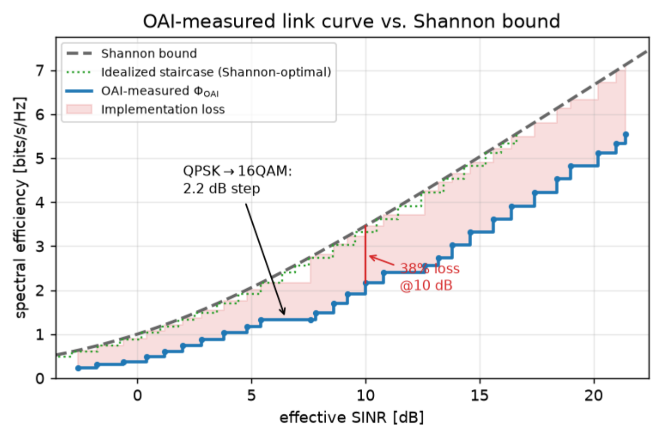

Fig. 2. OAI-measured link curve Φ_{OAI} vs. Shannon bound and an idealized (Shannon-optimal) staircase. The shaded region is the implementation loss (38% at 10 dB). Non-uniform MCS step spacing reflects a real LDPC decoder.

On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s}\,\eta_u\,(1 - \mathrm{BLER}_{\mathrm{tgt}}), \tag{4}$$

where B_s is the serving cell bandwidth and K_s its number of attached UEs.

V. Traffic-Steering rApp

The rApp adjusts the CIO vector to maximise the proportional-fair utility $U(\mathrm{CIO}) = \sum_u \log R_u$ (Alg. 1). Because U is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus; we therefore line-search each cell's CIO over a grid (coordinate ascent), which is monotonically non-decreasing in U by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

Every KPI evaluation reuses the same engine with a `cio_db` vector, so the control loop rides on the exact same twin used for the baseline. The rApp's output is a set of per-UE handover decisions that a real stack would execute via

Algorithm 1 Proportional-fair traffic steering

1) CIO $\leftarrow \mathbf{0}$; evaluate R_u and U via Eqs. (1)–(4).
2) **repeat** (sweep over cells):
    a) for each cell c: line-search CIO_c over a grid (others fixed) and keep the value maximising U.
3) **until** no cell's line-search improves U (local optimum).

O1/telnet. The coordinate-ascent structure guarantees monotonic improvement in U and converges in 2–3 sweeps, suitable for the non-RT RIC timescale.

VI. Twin-to-Real-Stack Bridges

Two bridges connect the analytical twin to a live OAI stack, grounding per-link realism.

Channel injection. A bridge converts a link's ray-traced channel frequency response to a sample-spaced impulse response (IFFT), retains the strongest causal taps, normalises them to unit energy, and maps each tap delay to a sample index. These taps override the rfsimulator's channel through a patch to OAI's channel module (`oai_rt_inject_channel` in `random_channel.c`). When `OAI_RT_TAPS` is set, the gNB–UE link runs over the site-specific channel. Injection is confirmed in the gNB log (`[RT] injected 4 taps ... channel_length=49`); the link then runs its PHY/MAC over the ray-traced channel.

Interference-as-noise-floor. A live OAI rfsimulator link is single-cell. A second bridge computes, per UE, the inter-cell interference and expresses it as a noise-floor rise $\Delta N_u = 10\log_{10}((I_u + N_0)/N_0)$, where I_u is the load-scaled sum of non-serving cells' received power. This is written as a per-client OAI noise power, so a single-link run reflects its multi-cell context—a site-specific generalisation of OCNG [9].

VII. Experimental Setup

Scenes. Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry fetched/built once per scene and reused across seeds.

Random layouts. Eight seeded layouts per scene—24 runs (323 s on one CPU server). Radio and topology parameters are summarised in Table II.

Configuration sweep. Neighbour load factor in $\{0, 0.25, 0.5, 0.75, 1.0\}$ on a cached channel per scene.

Metrics. We report per-UE downlink throughput (median and mean); the *served cell-edge* rate, defined as the 5th percentile among UEs with non-zero throughput (robust to coverage holes, unlike a raw 5th percentile which collapses to zero once outage exceeds 5%); the *outage rate*, the fraction of UEs with zero throughput; Jain's fairness index over per-UE rates; and the peak cell load (largest number of UEs on any cell). rApp gains are reported as mean ± standard deviation over the eight seeds per scene, and pooled across all 24 runs; a gain is undefined (excluded) for a run whose baseline metric is zero.

TABLE II
Radio and topology parameters.

Parameter	Value
Carrier / bandwidth	3.5 GHz / 20 MHz
Subcarrier spacing	30 kHz (numerology 1)
Cell transmit power	46 dBm
BS array / UE antennas	4×1 / 1
Receiver noise figure	7 dB
Ray-tracing interaction depth	3
Cells / UEs per layout	9 / 30
Default inter-cell load ρ	1.0

TABLE III
rApp gains vs. strongest-cell (mean ± std over seeds).

Metric	Overall	Berlin	Paris	Manh.
Median tput gain	$9.5 \pm 11.3\%$	10.5%	4.5%	13.3%
Served edge gain	$22.4 \pm 24.8\%$	41.0%	6.0%	20.2%
Jain fairness gain	$9.1 \pm 11.0\%$	11.2%	4.4%	11.5%
Peak load (UEs)	−0.75	−0.88	−0.50	−0.88
Mean tput gain	$0.5 \pm 5.8\%$	−0.7%	−0.3%	2.5%
Outage change	0.0 pts	0.0	0.0	0.0

VIII. Results

A. Traffic-steering rApp

Across 24 runs, relative to strongest-cell association (Table III, Fig. 3): the rApp helps the users a strongest-cell policy under-serves (median, cell-edge, fairness) and sheds peak load. In absolute terms the overall median per-UE rate rises from 8.5 to 9.2 Mbps and the served cell-edge rate from 3.5 to 4.2 Mbps, while the mean is essentially unchanged (11.4 Mbps). Three honest observations: aggregate rate barely moves (+0.5%; the proportional-fair objective trades sum for fairness), outage is unchanged (steering relocates covered users, it does not create coverage), and variance is high (layout-dependent)—motivating multi-seed evaluation.

The per-scene breakdown is instructive (Table III). Berlin, the densest scene, shows the largest served cell-edge gain (+41%): dense blockage produces stronger load imbalance for the rApp to exploit. Paris shows the smallest gain (+6%) and the highest outage (23.8%): when many users lack coverage entirely, steering has little covered load to rebalance. Manhattan sits in between. Across all three cities the direction is robust—median, cell-edge, and fairness improve while peak load falls—while the magnitude is strongly deployment-dependent, exactly what a single-scenario study would misrepresent.

B. The fidelity gap changes rApp evaluation

The 38% link-curve gap of Sec. IV is not cosmetic—it changes what the twin tells an rApp developer. We re-run the *identical* rApp on a twin whose *only* change is the link curve: Φ_{OAI} is replaced by an idealized curve that assigns the same MCS set at the Shannon-optimal SINR (no implementation loss), with the ray-traced channel, association, and interference held fixed. As expected, the idealized twin inflates baseline median throughput by 35%. More consequentially, it *disagrees*

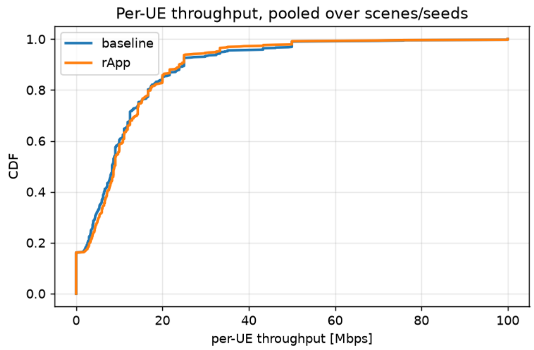

Fig. 3. Per-UE throughput CDF, baseline vs. rApp (pooled over all scenes and seeds). The rApp lifts the low-to-median part of the distribution.

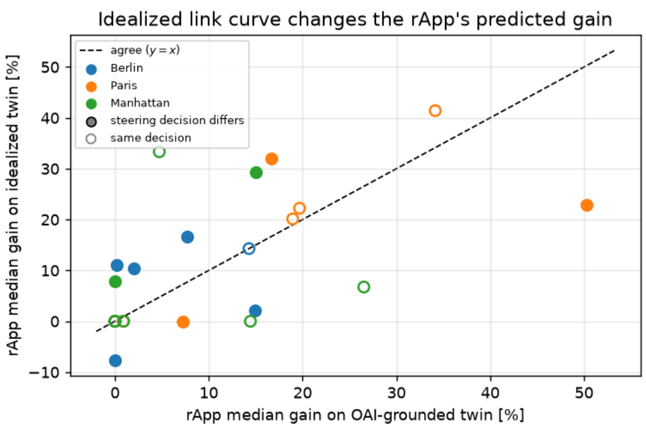

Fig. 4. Predicted rApp median gain on an idealized-curve twin vs. the OAI-grounded twin (24 layouts). Points off $y=x$ show the idealized twin mis-estimating the gain; filled markers are the 10/24 layouts where it also selects different steering actions.

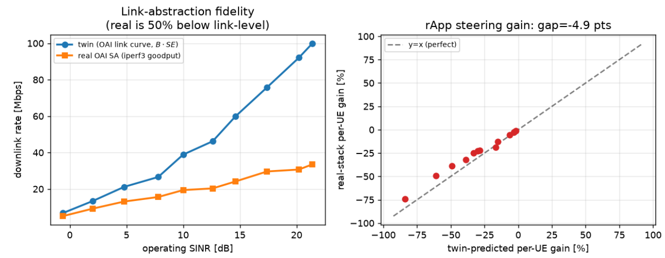

Fig. 5. Measured real fidelity gap. Left: twin link-level rate vs. real OAI Standalone end-to-end goodput (iperf3) at matched operating SINR—real is a median 53% lower. Right: per-UE rApp steering gain, twin-predicted vs. real-stack (8 layouts); points below $y=x$ show the twin overstating each move (gap 4.9 pts).

with the real-PHY twin on the control decision: across the 24 layouts its predicted rApp median gain deviates from the OAI-grounded value by a median of 7.5 (mean 8.2, max 29) points, the two agree only moderately (r=0.59; Fig. 4), and in 10/24 layouts (42%) the rApp selects a *different* set of steering actions (mean steered-user Jaccard 0.75). Yet the *ensemble-average* gain is nearly identical (10.9% vs 10.4%): an idealized twin looks calibrated in aggregate while being unreliable per deployment—exactly the regime in which an operator tunes and ranks a policy for a specific site. This is the concrete cost of the fidelity gap, and why a control-oriented twin must be grounded in a real receiver.

C. The real fidelity gap

The analysis above compares two *models*; we now close the loop on the real stack and measure the gap directly. We stand up a full OAI 5G Standalone system on the same CPU host—the OAI 5G core (AMF/SMF/UPF and supporting NFs, all healthy), a gNB, and a real UE that registers, establishes a PDU session (DL IP over the UPF), and carries user-plane traffic—and confirm end-to-end connectivity by iperf3 to the data network. Because the rfsimulator is single-gNB, we ground the *link* dimension of the rApp decision on the real stack: we pin the downlink operating point by capping the scheduler MCS over a clean channel, sweep the cap across MCS 2–28, and measure end-to-end iperf3 goodput at each fixed MCS, which indexes the same OAI curve the twin uses (a shared SINR axis). The rApp's load-balancing dimension remains evaluated on the multi-cell analytical twin (Sec. VIIIA).

The result is Fig. 5. The twin's link-level rate $B\,\eta_{\mathrm{OAI}}(1-\mathrm{BLER})$ overpredicts real end-to-end goodput by a median of 53% (mean 50%)—e.g. at MCS 28, 100 vs. 33.5 Mbps. This gap is *larger* than the link-model gap of Sec. IV and distinct in origin: Φ_{OAI} is measured on an isolated, full-buffer `nr_dlsim` link, whereas the real stack pays for the TDD frame structure (7D:2U), HARQ, RLC/PDCP/SDAP/GTP headers, scheduler dynamics, and TCP. Propagated through the rApp over eight layouts (240 per-UE evaluations, 13 steered UEs), the twin-predicted per-UE steering gain (−28.5% mean) and the real gain (−23.7%) differ by 4.9 ± 4.3 points: the real stack's flatter rate-vs-SINR curve compresses every move, consistently overstated by the twin. A policy tuned on link-level rates misjudges its own impact until closed against the real stack.

D. Coverage and outage

Mean UE outage is 16.2% at full load, and scene-dependent (Berlin 8.8%, Manhattan 16.3%, Paris 23.8%), reflecting random placement over the extent. For a fixed layout, outage is coverage-limited (path-loss-driven) and insensitive to inter-cell load, while served users are interference-limited. The rApp gain tracks this coverage regime: across runs, median gain declines with baseline outage (r=−0.42), whereas raw load imbalance (peak-to-mean load) is only weakly predictive (R^2=0.04)—steering helps where there is *covered* load to rebalance, not merely where load is uneven.

E. Sensitivity to inter-cell load

As the neighbour load factor falls from 1.0 to 0, mean per-UE throughput rises 1.7–1.8× (Berlin 13.6 → 22.7, Paris 14.9 → 25.2, Manhattan 11.9 → 22.0 Mbps), confirming served users are interference-limited. Combined with the outage result, this separates two regimes the twin must represent distinctly: a coverage-limited tail, set by path loss and insensitive to load, that steering cannot fix; and an interference-limited body, where both the neighbour load and the rApp's load-balancing act. A twin reporting only average throughput would blur these regimes.

F. Real-stack grounding

The channel-injection bridge (Sec. VI) is verified on a live OAI gNB–UE link on the same CPU host: injection is confirmed in the gNB log (`[RT] injected 4 taps ...channel_length=49`) and the link runs its PHY/MAC (HARQ, link adaptation) over the site-specific channel. Absolute goodput then comes from the full-stack Standalone path measured in Sec. VIII-C, where the OAI 5G core, gNB, a real UE PDU session, and iperf3 over the UPF form a running loop—what makes the fidelity gap measurable rather than projected.

IX. Discussion and Limitations

We state these plainly. (1) The full OAI Standalone loop runs and the real fidelity gap is measured (Sec. VIII-C), but on a single gNB: the rApp's inter-cell load-balancing is evaluated on the analytical twin, while the real stack grounds the link/operating-point dimension; a single-run, multi-gNB N2 handover driven by the rApp is the remaining step. (2) Inter-cell interference is modelled as a noise-floor rise, not physically combined waveforms (which need FPGA/GPU emulators). (3) The evaluation is a static full-buffer snapshot, single carrier, downlink; mobility, FFR, and uplink are natural extensions. (4) rApp benefits are high-variance and layout-dependent (we report full distributions), and steering does not address coverage. (5) Validation is stack-versus-model; field calibration and real cell-site databases are the next steps.

X. Conclusion

We presented an open, CPU-only, site-specific, multi-cell 5G digital twin that grounds throughput in OAI's physical layer—replacing idealized link-to-system curves with a curve measured from a real 5G receiver—and hosts a traffic-steering rApp. We quantified the implementation loss (38% below Shannon at 10 dB) and, crucially, showed it is first-order for control: an idealized-curve twin changes the rApp's steering decisions in 42% of layouts and mis-estimates its per-deployment gain by a median of 7.5 points, so link-model fidelity—not only channel fidelity—determines whether a twin can be trusted to tune a policy. Two bridges connect the analytical twin to a live OAI stack: exact ray-traced channel injection and a per-UE interference-as-noise-floor mapping. Across three cities and eight random layouts each, the rApp improves median and cell-edge throughput and fairness and relieves peak load, without inflating aggregate rate or fixing coverage—an honest, reproducible result. The rApp gain is limited by baseline outage ($r{=}-0.42$) rather than by raw load imbalance, clarifying where steering helps. Finally, we closed the loop on a live OAI 5G Standalone stack—5G core, gNB, a real UE PDU session, and iperf3—on the same commodity host and *measured* the fidelity gap: deliverable end-to-end goodput is a median 53% below the twin's link-level prediction, and the rApp's per-UE steering gain is mis-estimated by 4.9 points. Link-model fidelity is thus necessary but not sufficient: even an OAI-grounded twin must be closed against the real stack before its control conclusions can be trusted.

Future work. A single-run, multi-gNB closed loop—live N2 handovers driven by the rApp with FlexRIC [13] KPM streaming—would extend the measured gap to the load-balancing dimension. Beyond that, real cell-site databases, mobility, uplink and multi-carrier support, and calibration against field measurements are the natural directions; each slots into the existing pipeline without changing its structure.

Reproducibility

An open harness reproduces the 24-run matrix and load sweep on a commodity CPU in $\sim$5 minutes; `experiments/real_fidelity_gap.py` drives the live OAI stack for the measured gap. The twin, the OAI-measured curve Φ_{OAI}, the rApp, the bridges, and the OAI scripts are released together under Apache-2.0.

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